

## SAFETY DATA SHEET

Creation Date 09-Nov-2010

Revision Date 21-Jun-2023

Revision Number 6

### 1. Identification

**Product Name** Ammonium bicarbonate

**Cat No. :** AC393210000; AC393210010; AC393210050; AC393212500

**CAS No** 1066-33-7  
**Synonyms** Ammonium hydrogen carbonate

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

**Company**

Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

**Emergency Telephone Number**

For information **US** call: 001-800-ACROS-01 / **Europe** call: +32 14 57 52 11  
**Emergency Number US:**001-201-796-7100 / **Europe:** +32 14 57 52 99  
**CHEMTREC** Tel. No.**US:**001-800-424-9300 / **Europe:**001-703-527-3887

### 2. Hazard(s) identification

**Classification**

This chemical is considered hazardous by the 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

Acute oral toxicity

Category 4

**Label Elements****Signal Word**

Warning

**Hazard Statements**

Harmful if swallowed

**Precautionary Statements****Prevention**

Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product

**Ingestion**

IF SWALLOWED: Call a POISON CENTER or doctor/physician if you feel unwell  
Rinse mouth

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Hazards not otherwise classified (HNOC)**

None identified

### 3. Composition/Information on Ingredients

| Component                  | CAS No    | Weight % |
|----------------------------|-----------|----------|
| Ammonium bicarbonate (1:1) | 1066-33-7 | >95      |

### 4. First-aid measures

|                                            |                                                                                                                   |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                      | If symptoms persist, call a physician.                                                                            |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.   |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician. |
| <b>Inhalation</b>                          | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.      |
| <b>Ingestion</b>                           | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.             |
| <b>Most important symptoms and effects</b> | None reasonably foreseeable.                                                                                      |
| <b>Notes to Physician</b>                  | Treat symptomatically                                                                                             |

### 5. Fire-fighting measures

|                                       |                                                                                       |
|---------------------------------------|---------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>   | Substance is nonflammable; use agent most appropriate to extinguish surrounding fire. |
| <b>Unsuitable Extinguishing Media</b> | No information available                                                              |
| <b>Flash Point</b>                    | No information available                                                              |
| <b>Method -</b>                       | No information available                                                              |
| <b>Autoignition Temperature</b>       | No information available                                                              |
| <b>Explosion Limits</b>               | No information available                                                              |

|                                         |                          |
|-----------------------------------------|--------------------------|
| <b>Upper</b>                            | No data available        |
| <b>Lower</b>                            | No data available        |
| <b>Sensitivity to Mechanical Impact</b> | No information available |
| <b>Sensitivity to Static Discharge</b>  | No information available |

**Specific Hazards Arising from the Chemical**

Non-combustible, substance itself does not burn but may decompose upon heating to produce corrosive and/or toxic fumes.

**Hazardous Combustion Products**

Ammonia.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
2

**Flammability**  
0

**Instability**  
1

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                             |                                                                                                              |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation.            |
| <b>Environmental Precautions</b>            | Do not flush into surface water or sanitary sewer system.                                                    |
| <b>Methods for Containment and Clean Up</b> | Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal. |

## 7. Handling and storage

|                 |                                                                                                                                                                                     |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid dust formation. Avoid ingestion and inhalation. Do not get in eyes, on skin, or on clothing. |
| <b>Storage.</b> | Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Incompatible Materials. Strong acids. Alkaline.                                                       |

## 8. Exposure controls / personal protection

|                                      |                                                                                                                                                                                                                                                   |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure Guidelines</b>           | This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies.                                                                                                     |
| <b>Engineering Measures</b>          | Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.                                                                                                 |
| <b>Personal Protective Equipment</b> |                                                                                                                                                                                                                                                   |
| <b>Eye/face Protection</b>           | Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.                                                                       |
| <b>Skin and body protection</b>      | Wear appropriate protective gloves and clothing to prevent skin exposure.                                                                                                                                                                         |
| <b>Respiratory Protection</b>        | Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. |
| <b>Recommended Filter type:</b>      | Particulates filter conforming to EN 143.                                                                                                                                                                                                         |
| <b>Hygiene Measures</b>              | Handle in accordance with good industrial hygiene and safety practice.                                                                                                                                                                            |

## 9. Physical and chemical properties

|                                        |                                   |
|----------------------------------------|-----------------------------------|
| Physical State                         | Powder Solid                      |
| Appearance                             | White                             |
| Odor                                   | Ammonia-like                      |
| Odor Threshold                         | No information available          |
| pH                                     | 8 50 g/l H <sub>2</sub> O         |
| Melting Point/Range                    | °C / 221 °F                       |
| Boiling Point/Range                    | No information available          |
| Flash Point                            | No information available          |
| Evaporation Rate                       | Not applicable                    |
| Flammability (solid,gas)               | No information available          |
| Flammability or explosive limits       |                                   |
| Upper                                  | No data available                 |
| Lower                                  | No data available                 |
| Vapor Pressure                         | 67 hPa @ 20 °C                    |
| Vapor Density                          | Not applicable                    |
| Specific Gravity                       | No information available          |
| Solubility                             | No information available          |
| Partition coefficient; n-octanol/water | No data available                 |
| Autoignition Temperature               | No information available          |
| Decomposition Temperature              | 60 °C                             |
| Viscosity                              | Not applicable                    |
| Molecular Formula                      | C H <sub>5</sub> N O <sub>3</sub> |
| Molecular Weight                       | 79.06                             |

## 10. Stability and reactivity

|                                  |                                                                                                |
|----------------------------------|------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                                     |
| Stability                        | Stable under normal conditions. heat sensitive.                                                |
| Conditions to Avoid              | Avoid dust formation. Incompatible products. Heat, flames and sparks. Temperatures above 35°C. |
| Incompatible Materials           | Strong acids, Alkaline                                                                         |
| Hazardous Decomposition Products | Ammonia                                                                                        |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                       |
| Hazardous Reactions              | None under normal processing.                                                                  |

## 11. Toxicological information

### Acute Toxicity

#### Product Information Component Information

| Component                  | LD50 Oral                 | LD50 Dermal                  | LC50 Inhalation |
|----------------------------|---------------------------|------------------------------|-----------------|
| Ammonium bicarbonate (1:1) | LD50 = 1576 mg/kg ( Rat ) | LD50 > 5000 mg/kg ( Rabbit ) | Not listed      |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | No information available                                                                 |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component                  | CAS No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|----------------------------|-----------|------------|------------|------------|------------|------------|
| Ammonium bicarbonate (1:1) | 1066-33-7 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Contains a substance which is: Very toxic to aquatic organisms. The product contains following substances which are hazardous for the environment.

| Component                  | Freshwater Algae | Freshwater Fish                                | Microtox                                      | Water Flea             |
|----------------------------|------------------|------------------------------------------------|-----------------------------------------------|------------------------|
| Ammonium bicarbonate (1:1) | Not listed       | LC50: 63.4 mg/L (96h)<br>(Oncorhynchus mykiss) | EC10: 1347 mg/L (16h)<br>(Pseudomonas putida) | EC50: 145.6 mg/L (48h) |

**Persistence and Degradability** Persistence is unlikely

**Bioaccumulation/ Accumulation** No information available.

**Mobility** . Will likely be mobile in the environment due to its water solubility.

| Component                  | log Pow |
|----------------------------|---------|
| Ammonium bicarbonate (1:1) | -2.4    |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

**DOT** Not regulated  
**TDG** Not regulated  
**IATA** Not regulated  
**IMDG/IMO** Not regulated

## 15. Regulatory information

### United States of America Inventory

| Component                  | CAS No    | TSCA | TSCA Inventory notification - Active-Inactive | TSCA - EPA Regulatory Flags |
|----------------------------|-----------|------|-----------------------------------------------|-----------------------------|
| Ammonium bicarbonate (1:1) | 1066-33-7 | X    | ACTIVE                                        | -                           |

**Legend:**

TSCA US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

X - Listed

- - Not Listed

**TSCA - Per 40 CFR 751, Regulation of Certain Chemical Substances & Mixtures, Under TSCA Section 6(h) (PBT)**

Not applicable

**TSCA 12(b)** - Notices of Export

Not applicable

**International Inventories**

Canada (DSL/NDSL), Europe (EINECS/ELINCS/NLP), Philippines (PICCS), Japan (ENCS), Japan (ISHL), Australia (AICS), China (IECSC), Korea (KECL).

| Component                  | CAS No    | DSL | NDSL | EINECS    | PICCS | ENCS | ISHL | AICS | IECSC | KECL     |
|----------------------------|-----------|-----|------|-----------|-------|------|------|------|-------|----------|
| Ammonium bicarbonate (1:1) | 1066-33-7 | X   | -    | 213-911-5 | X     | X    | X    | X    | X     | KE-01678 |

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)**U.S. Federal Regulations****SARA 313**

| Component                  | CAS No    | Weight % | SARA 313 - Threshold Values % |
|----------------------------|-----------|----------|-------------------------------|
| Ammonium bicarbonate (1:1) | 1066-33-7 | >95      | 1.0                           |

**SARA 311/312 Hazard Categories** See section 2 for more information**CWA (Clean Water Act)**

| Component                  | CWA - Hazardous Substances | CWA - Reportable Quantities | CWA - Toxic Pollutants | CWA - Priority Pollutants |
|----------------------------|----------------------------|-----------------------------|------------------------|---------------------------|
| Ammonium bicarbonate (1:1) | X                          | 5000 lb                     | -                      | -                         |

**Clean Air Act**

Not applicable

**OSHA** - Occupational Safety and Health Administration

Not applicable

**CERCLA**

This material, as supplied, contains one or more substances regulated as a hazardous substance under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302)

| Component                  | Hazardous Substances RQs | CERCLA EHS RQs |
|----------------------------|--------------------------|----------------|
| Ammonium bicarbonate (1:1) | 5000 lb                  | -              |

**California Proposition 65**

This product does not contain any Proposition 65 chemicals.

**U.S. State Right-to-Know Regulations**

| Component                  | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|----------------------------|---------------|------------|--------------|----------|--------------|
| Ammonium bicarbonate (1:1) | X             | X          | X            | -        | -            |

**U.S. Department of Transportation**

Reportable Quantity (RQ): Y

DOT Marine Pollutant N

DOT Severe Marine Pollutant N

## U.S. Department of Homeland Security

This product does not contain any DHS chemicals.

Other International Regulations

## Mexico - Grade

No information available

## Authorisation/Restrictions according to EU REACH

| Component                  | CAS No    | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|----------------------------|-----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Ammonium bicarbonate (1:1) | 1066-33-7 | -                                                                   | Use restricted. See item 65.<br>(see link for restriction details)            | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                  | CAS No    | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|----------------------------|-----------|----------|------------------------------|---------------------------|--------------------------------------------|
| Ammonium bicarbonate (1:1) | 1066-33-7 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component                  | CAS No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|----------------------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Ammonium bicarbonate (1:1) | 1066-33-7 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

**16. Other information**

## Prepared By

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09-Nov-2010

## Revision Date

21-Jun-2023

## Print Date

21-Jun-2023

## Revision Summary

This document has been updated to comply with the US OSHA HazCom 2012 Standard replacing the current legislation under 29 CFR 1910.1200 to align with the Globally Harmonized System of Classification and Labeling of Chemicals (GHS).

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**